

# A32NX flyPad failures

Clearer A32NX failure names and new failures with their FCOM alerts: transponders, RMPs and VHF radios, FWC 1 and 2, engine and APU fires with a real fire detection model and fire detection unit failures, wing tank pumps, the X FEED valve, the centre tank transfer valves, engine flameouts and seizures with the in-flight relight envelope, and the engine oil, ignition and start, stall, EGT, overspeed, vibration, FADEC, thrust lever and reverser failures with the FCOM start sequence, the fan blocked failure with ENG LOW N1, and the THR LEVER ABV IDLE and SAT ABOVE FLEX TEMP alerts

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| Aircraft         | A32NX                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Where to find it | flyPad > Failures (ATA 21, 22, 23, 26, 28, 31, 34, 72, 73, 74, 77, 78, 79, 80); E/WD, STATUS and SD pages; pedestal ATC and RMP panels; overhead FIRE, FUEL and ENG MAN START panels; ENG MASTER and ENG MODE controls; thrust levers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Branch           | core/failures-a32nx, based on core/gps (stacked: needs core/surveillance, core/a380x-mfd-fms and core/gps merged first), with core/fire merged in (the fire failures use its fire warnings, and the engine failures use its ENG FIRE pb LP valve starvation). Only the seventeen features below belong to this branch; the GPS and fire warning features are described in the core/gps and core/fire manuals. Some engine commits also carry A380X files (they are on core/failures-a380x ); this manual describes only what the A32NX gets.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Status           | Transponder, RMP/VHF, FWC, fire and fuel failures tested in the sim by the user: OK (2026-10-04); the fire detection unit failures, the FIRE TEST light logic, the APU fire horn and the centre tank transfer valve failures too (2026-10-04, the horn itself is silent in MSFS 2024); the engine flameout and seizure failures on develop: OK, tests T1-T9 (2026-10-05); their LEAP-1A relight envelope: sim-tested OK 2026-10-05 (recorded); the in-flight start without N2 overshoot: OK on develop (2026-10-06, recorded). Engine failures A2-A5 on develop (2026-10-06 integration flight and 2026-10-07 retest, recorded): oil failures, ground starts with a failed igniter and the dry crank, stall, EGT overtemperature and vibration, thrust lever resolvers and FADEC FAULT, normal reverse thrust: OK; the other engine failures, the later tuning and the T.O CONFIG NORMAL / WING TK LO LVL fixes: not sim-tested. Clearer names: not sim-tested (labels only). Engine alerts 2026-10-08 (feature branch): SAT ABOVE FLEX TEMP OK (recorded); THR LEVER ABV IDLE and the fan blocked / LOW N1 not sim-tested yet. The fuel filter clog and the stall during a relight fix: not sim-tested. Nothing is sim-tested on this branch itself. |
| Pictures         | No pictures: the failures page, the ECAM alerts and the panels are the stock FBW displays.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

1. Clearer failure names

2. Transponder failures

3. RMP and VHF failures

4. FWC 1 and FWC 2 failures

5. Fire detection model and fire failures

6. Fire detection unit failures, FIRE TEST light logic, APU fire horn

7. Wing tank pump and X FEED valve failures

8. Centre tank transfer valve failures and fuel follow-ups

9. Engine flameout and seizure failures, in-flight relight
10. Engine oil failures: leak, filter clog, overheat

11. Engine fuel filter clog

12. Engine ignition and start failures, FCOM start sequence

13. Engine stall, EGT overtemperature, overspeed and vibration failures

14. FADEC, thrust lever and reverser failures

15. T.O CONFIG NORMAL and FUEL WING TK LO LVL fixes

16. ENG THR LEVER ABV IDLE and ENG SAT ABOVE FLEX TEMP

17. Engine fan blocked failure and ENG LOW N1

## 1. Clearer failure names

|           |                               |
|-----------|-------------------------------|
| Aircraft: | A32NX                         |
| Where:    | flyPad > Failures             |
| Status:   | Not sim-tested (labels only). |

### What it is for

The failure names were audited ATA by ATA against what the systems code fails. Three names were misleading: the ATA 21 valve is the hot air pressure regulating valve, the FMGC failures only fail the flight guidance part of the FMGC,

and the RA failures are antenna failures. Only the labels change.

| ATA | Before                                      | Now                                                         |
|-----|---------------------------------------------|-------------------------------------------------------------|
| 21  | Trim Air Pressure Regulating Valve          | Hot Air Pressure Regulating Valve                           |
| 22  | FMGC 1 / FMGC 2                             | FMGC 1 (flight guidance) / FMGC 2 (flight guidance)         |
| 34  | RA 1 Interrupted / RA 2 Interrupted         | RA 1 antenna interrupted / RA 2 antenna interrupted         |
| 34  | RA 1 Direct Coupling / RA 2 Direct Coupling | RA 1 antenna direct coupling / RA 2 antenna direct coupling |

## How to use

1. Open flyPad > Failures and pick the failure by its new name (table above).

## How to test

| # | Do this                                           | Expected result                                                                                                         |
|---|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| 1 | Open flyPad > Failures > ATA 21 Air Conditioning. | "Hot Air Pressure Regulating Valve" is listed.                                                                          |
| 2 | Open ATA 22 Auto Flight.                          | "FMGC 1 (flight guidance)" and "FMGC 2 (flight guidance)".                                                              |
| 3 | Open ATA 34 Navigation.                           | "RA 1 antenna interrupted", "RA 2 antenna interrupted", "RA 1 antenna direct coupling", "RA 2 antenna direct coupling". |

## 2. Transponder failures

**Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 34 Navigation; pedestal ATC/TCAS panel; E/WD, STATUS | **Status:** Tested in the sim by the user: OK (2026-10-04).

## What it is for

Two new failures, **ATC/XPDR 1** and **ATC/XPDR 2**. MSFS has a single transponder: it stands for the transponder selected on the ATC panel (SYS 1 / SYS 2). When the selected one fails, the sim transponder stops replying and the ATC FAIL light comes on; selecting the other system brings the replies back. With both lost, the TCAS goes to standby. The ECAM alerts and STATUS follow the A320 FCOM PRO-ABN-NAV procedures.

Fix on the way: the ATC/XPDR STBY indication read a second sim transponder that does not exist; the standby state of each transponder now comes from the system selection and the mode selector.

## What you should see

| Situation                         | What you should see                                                                                                                                                                                  |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ATC/XPDR 1 failed, SYS 1 selected | ATC FAIL light; no transponder reply. E/WD amber <b>NAV ATC/XPDR 1 FAULT</b> with -ATC/XPDR ..... SYS 2 (only while XPDR 1 is selected and XPDR 2 works). STATUS INOP SYS: ATC/XPDR 1, ADS-B RPTG 1. |
| SYS 2 selected                    | ATC FAIL off, replies again; the procedure line is done.                                                                                                                                             |
| ATC/XPDR 2 failed                 | The same with NAV ATC/XPDR 2 FAULT, -ATC/XPDR ..... SYS 1, INOP SYS ATC/XPDR 2, ADS-B RPTG 2.                                                                                                        |
| Both failed                       | <b>NAV ATC/XPDR 1+2 FAULT</b> (crew awareness); STATUS INOP SYS: TCAS, ATC/XPDR 1, ATC/XPDR 2, ADS-B RPTG 1, ADS-B RPTG 2; the TCAS goes to standby.                                                 |

A transponder loses its bus (AC ESS SHED for XPDR 1, AC 2 for XPDR 2)      The transponder is lost, but no ATC/XPDR FAULT: a bus loss is an ELEC alert.

## Limitations and why

| Limitation                                                                                             | Why                                                                                                        | What to do                     |
|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|--------------------------------|
| The ATC/XPDR alerts are inhibited during the takeoff and below 800 ft in approach to the landing roll. | The FCOM gives the inhibition as a figure: the one of the RA faults is used (flight phases 3, 4, 5, 7, 8). | Test at the gate or in cruise. |

## How to use

1. Open flyPad > Failures > ATA 34 Navigation and activate **ATC/XPDR 1** or **ATC/XPDR 2**.
2. Follow the ECAM: select the other system on the ATC/TCAS panel.

## How to test

| # | Do this                                                               | Expected result                                                                                                                                                    |
|---|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | A32NX at the gate, powered, ATC/TCAS panel: SYS 1, mode AUTO (or ON). | No ATC FAIL light.                                                                                                                                                 |
| 2 | Activate <b>ATC/XPDR 1</b> .                                          | ATC FAIL light on; master caution and single chime; E/WD NAV ATC/XPDR 1 FAULT with -ATC/XPDR ..... SYS 2. ATC (or a traffic tool) no longer sees your transponder. |
| 3 | Open the STATUS page.                                                 | INOP SYS: ATC/XPDR 1, ADS-B RPTG 1.                                                                                                                                |
| 4 | Select SYS 2 on the ATC panel.                                        | ATC FAIL off, the transponder replies again, the SYS 2 line is done.                                                                                               |
| 5 | Activate <b>ATC/XPDR 2</b> as well.                                   | NAV ATC/XPDR 1+2 FAULT; STATUS INOP SYS: TCAS, ATC/XPDR 1, ATC/XPDR 2, ADS-B RPTG 1, ADS-B RPTG 2; the TCAS is in standby.                                         |
| 6 | Deactivate both failures.                                             | The alerts clear; the TCAS works again.                                                                                                                            |

## 3. RMP and VHF failures

**Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 23 Communication; pedestal RMP 1 and RMP 2 | **Status:** Tested in the sim by the user: OK (2026-10-04).

## What it is for

Five new ATA 23 failures: **RMP 1**, **RMP 2**, **VHF 1**, **VHF 2** and **VHF 3** (same ids as the A380X ATA 23 failures). A failed RMP goes blank and stops tuning; the other RMP still tunes every VHF (A320 FCOM DSC-23-10-20: "If one RMP fails, the remaining one controls all the VHF transceivers"). A failed VHF switches its MSFS COM radio off: it neither receives nor transmits.

## What you should see

| Situation           | What you should see                                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------|
| RMP 1 (or 2) failed | Both frequency displays blank, the green key lights and SEL off, the keys and knobs do nothing (FCOM DSC-23-60 failure cases). |

|                     |                                                                                                   |
|---------------------|---------------------------------------------------------------------------------------------------|
| The other RMP       | Still tunes VHF 1, 2 and 3.                                                                       |
| VHF 1 (2, 3) failed | COM 1 (2, 3) silent: no reception, no transmission. The RMPs still show and change its frequency. |

### Limitations and why

| Limitation                                                            | Why                                                                                           | What to do                               |
|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|------------------------------------------|
| No ECAM alert for these failures.                                     | Not part of this change.                                                                      | Check the RMP and the radio.             |
| A NAV key left on on a failed RMP still holds the FMGC navaid tuning. | As with an RMP switched off: the navaid tuner (shared with the A380X) reads the NAV key only. | Avoid the NAV key before failing an RMP. |
| No RMP 3 and no audio control panel / AMU failure.                    | RMP 3 is not in the A32NX cockpit, and the MSFS radios have no audio channel to fail.         | None.                                    |

### How to use

1. Open flyPad > Failures > ATA 23 Communication and activate RMP 1, RMP 2, VHF 1, VHF 2 or VHF 3.
2. With an RMP failed, tune the radios from the other RMP.

### How to test

| # | Do this                                                                | Expected result                                                                                              |
|---|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| 1 | A32NX powered, an ATIS frequency on VHF 1, audible.                    | The ATIS is heard.                                                                                           |
| 2 | Activate <b>RMP 1</b> .                                                | RMP 1 goes blank, its key lights go off, its knobs do nothing. The ATIS is still heard (VHF 1 itself works). |
| 3 | On RMP 2, press the VHF 1 key and tune another frequency, transfer it. | VHF 1 changes frequency: RMP 2 tunes VHF 1.                                                                  |
| 4 | Deactivate RMP 1. Activate <b>VHF 1</b> .                              | COM 1 goes silent; transmitting on VHF 1 is not possible. RMP 1 still shows the frequency.                   |
| 5 | Deactivate VHF 1.                                                      | The ATIS is heard again.                                                                                     |

## 4. FWC 1 and FWC 2 failures

**Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 31 Indicating / Recording System; E/WD, STATUS | **Status:** Tested in the sim by the user: OK (2026-10-04).

### What it is for

Two new failures, **FWC 1** and **FWC 2** (A320 FCOM DSC-31-05-30, PRO-ABN-FWS). With one FWC failed, the other one keeps the warning system working and shows the fault. With both failed, the ECAM alerts, memos, STATUS, MASTER WARN / CAUT lights, aural (including the STALL aural) and the altitude callouts are lost, and the E/WD shows the FWC 1+2 FAULT display. Each FWC also follows its own electrical supply.

### What you should see

| Situation    | What you should see                                                                                                                      |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------|
| FWC 1 failed | E/WD <b>FWS FWC 1 FAULT</b> (shown by FWC 2), level 1: no MASTER CAUT, no chime. STATUS: CAT 3 SINGLE ONLY; INOP SYS: CAT 3 DUAL, FWC 1. |
| FWC 2 failed | The same with FWS FWC 2 FAULT and FWC 2.                                                                                                 |

|                  |                                                                                                                                                                                                                   |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Both failed      | E/WD: <b>FWS FWC 1+2 FAULT</b> , -MONITOR SYS, -MONITOR OVERHEAD PANEL; right side NOT AVAIL: ECAM WARN, ALTI ALERT, STATUS, A/CALL OUT, MEMO. No other alert, memo, STATUS, STS reminder, master light or aural. |
| One FWC restored | Every active alert is shown again, as a new one.                                                                                                                                                                  |

## Limitations and why

| Limitation                                                                                             | Why                                                                                                       | What to do                     |
|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|--------------------------------|
| FWS FWC 1(2) FAULT is level 1 and inhibited during the takeoff and the landing (phases 3, 4, 5, 7, 8). | The FCOM gives the level and the inhibition in figures only: those of the other computer faults are used. | Test at the gate or in cruise. |

## How to use

1. Open flyPad > Failures > ATA 31 Indicating / Recording System and activate FWC 1 and / or FWC 2.
2. With both failed, monitor the systems and the overhead panel yourself.

## How to test

| # | Do this                                                       | Expected result                                                                                                                                     |
|---|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | A32NX at the gate, powered, no alert. Activate <b>FWC 1</b> . | E/WD: FWS FWC 1 FAULT, no master caution, no chime.                                                                                                 |
| 2 | Open the STATUS page.                                         | CAT 3 SINGLE ONLY; INOP SYS: CAT 3 DUAL, FWC 1.                                                                                                     |
| 3 | Activate <b>FWC 2</b> as well.                                | E/WD: FWS FWC 1+2 FAULT, -MONITOR SYS, -MONITOR OVERHEAD PANEL, and NOT AVAIL: ECAM WARN, ALTI ALERT, STATUS, A/CALL OUT, MEMO. The memos are gone. |
| 4 | Press the STS key on the ECAM control panel.                  | No STATUS page (STATUS is in the NOT AVAIL list).                                                                                                   |
| 5 | Deactivate FWC 2.                                             | FWS FWC 1 FAULT again; the active alerts come back as new ones.                                                                                     |

## 5. Fire detection model and fire failures

**Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 26 Fire Protection; overhead FIRE panel; ENG MASTER panel; E/WD, STATUS | **Status:** Tested in the sim by the user: OK (2026-10-04).

### What it is for

Nine new failures: **Engine 1 fire, Engine 2 fire, APU fire**, and **Engine 1 fire detection loop A, Engine 1 fire detection loop B, Engine 2 fire detection loop A, Engine 2 fire detection loop B, APU fire detection loop A, APU fire detection loop B**.

Each zone now has the two detection loops and the fire detection unit of the aircraft (A320 FCOM DSC-26-20-10). A fire warning comes when both loops detect the fire, when one loop detects it and the other one is failed, or when both loops break within 5 s (flame effect). The logic moved from the A380X into the common code; the A380X behaviour is unchanged. The fires of the MSFS failures menu still give the warnings, through the loops. The FIRE TEST is unchanged.

- The ENG MASTER panel FIRE light comes on with the engine fire warning (before: FIRE TEST only).
- APU fire on the ground: the APU shuts down and the APU AGENT discharges automatically 3 s after the warning (FCOM DSC-26-20-20).

- Each AGENT discharge has one chance in two to put a flyPad fire out.
- Loop supplies (FCOM PRO-ABN-ELEC INOP SYS): DC BUS 2 for ENG 1 LOOP B and ENG 2 LOOP A, DC ESS BUS for ENG 1 LOOP A and ENG 2 LOOP B, DC BAT BUS for both APU loops. An unpowered loop is a failed loop.

## What you should see

| Situation                   | What you should see                                                                                                                                                                                   |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Engine 1 fire               | ENG 1 FIRE warning (from core/fire), red ENG 1 FIRE pushbutton light and red FIRE light on the ENG MASTER panel.                                                                                      |
| APU fire on the ground      | APU FIRE warning; the APU shuts down; 3 s later the APU AGENT discharges (DISCH light).                                                                                                               |
| AGENT discharged            | The fire goes out, or not (one chance in two per discharge).                                                                                                                                          |
| One loop failed             | Amber <b>ENG 1 FIRE LOOP A FAULT</b> (LOOP B, ENG 2, APU FIRE LOOP A / B); STATUS INOP SYS ENG 1 LOOP A (ENG 1 LOOP B, ENG 2 LOOP A / B, APU LOOP A / B). A fire is still detected by the other loop. |
| Both loops of a zone failed | <b>ENG 1 FIRE DET FAULT</b> (ENG 2, APU FIRE DET FAULT) instead of the two loop cautions; STATUS INOP SYS FIRE DET 1 (FIRE DET 2, APU FIRE DET).                                                      |

## Limitations and why

| Limitation                                                                                               | Why                                                                                           | What to do                        |
|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-----------------------------------|
| FIRE DET FAULT also comes from a failed fire detection unit.                                             | See the next feature, "Fire detection unit failures".                                         | -                                 |
| The LOOP and DET FAULT cautions are inhibited during the takeoff and the landing (phases 3, 4, 5, 7, 8). | The FCOM gives the inhibition as a figure: that of the other crew awareness cautions is used. | Test at the gate or in cruise.    |
| A full MSFS restart is needed after an update.                                                           | The systems WASM module and the cockpit behaviour XML change.                                 | Restart the sim after installing. |

## How to use

1. Open flyPad > Failures > ATA 26 Fire Protection and activate a fire or a loop failure.
2. Fight a fire with the ECAM procedure (FIRE pushbutton, AGENT 1, AGENT 2). Deactivating the failure also ends the fire.

## How to test

| # | Do this                                                                                             | Expected result                                                                                                             |
|---|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 1 | A32NX on the ground, engines running. Activate <b>Engine 1 fire</b> .                               | ENG 1 FIRE warning, master warning, continuous chime, red light in the ENG 1 FIRE pushbutton and on the ENG MASTER 1 panel. |
| 2 | Follow the ECAM: ENG MASTER 1 OFF, push the ENG 1 FIRE pushbutton, press AGENT 1.                   | AGENT 1 DISCH; the fire goes out or not (one chance in two). If not, AGENT 2 gives another chance.                          |
| 3 | Deactivate the failure, reset the FIRE pushbutton. With the APU running, activate <b>APU fire</b> . | APU FIRE warning; the APU shuts down; about 3 s later the APU AGENT discharges by itself.                                   |
| 4 | Deactivate it. Activate <b>Engine 2 fire detection loop A</b> .                                     | Amber ENG 2 FIRE LOOP A FAULT; STATUS INOP SYS ENG 2 LOOP A.                                                                |

|   |                                                                              |                                                                                     |
|---|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 5 | Activate <b>Engine 2 fire</b> with loop A still failed.                      | ENG 2 FIRE warning: loop B alone detects the fire.                                  |
| 6 | Deactivate the fire, activate <b>Engine 2 fire detection loop B</b> as well. | The loop cautions are replaced by ENG 2 FIRE DET FAULT; STATUS INOP SYS FIRE DET 2. |
| 7 | Optional: start an engine fire from the MSFS failures menu.                  | The ENG FIRE warning still comes (through the loops).                               |

## 6. Fire detection unit failures, FIRE TEST light logic, APU fire horn

**Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 26 Fire Protection; overhead FIRE panel; ENG MASTER panel; E/WD, STATUS | **Status:** Tested in the sim by the user: OK (2026-10-04); the horn sound itself is silent in MSFS 2024. Rust tests (fire\_protection), unit tests: FireDetectionFaults.spec.ts, EngineMasterFireLight.spec.ts.

### What it is for

Three follow-ups of the fire detection model above:

- **Fire detection unit failures:** three new failures, **Engine 1 fire detection unit (FDU)**, **Engine 2 fire detection unit (FDU)** and **APU fire detection unit (FDU)**. A failed FDU gives the FIRE DET FAULT caution of its zone even with healthy loops, and detects no fire. Clearing the failure with a fire present gives the fire warning at once.
- **ENG MASTER panel FIRE light during the FIRE TEST:** the A320 FCOM (DSC-26-20-20, ENG 1(2) FIRE TEST pb) says that the FIRE lights do not come on with an engine 1 loop A fault or an engine 2 loop B fault. The light now stays dark in those cases (a loop fault is its failure or the loss of its DC ESS supply).
- **APU fire external horn:** an APU fire on the ground sounds the horn in the nose gear bay (FCOM DSC-26-20-20 APU FIRE light, DSC-49-10-20), on the same sound as the MECH call; not during the FIRE TEST.

### What you should see

| Situation                                                  | What you should see                                                                                                                                                                                                             |
|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Engine 1 FDU failed                                        | Amber <b>ENG 1 FIRE DET FAULT</b> (ENG 2 FIRE DET FAULT, APU FIRE DET FAULT for the other units), no LOOP FAULT caution (a failed FDU cannot tell which loop is failed); STATUS INOP SYS FIRE DET 1 (FIRE DET 2, APU FIRE DET). |
| A fire in a zone with its FDU failed                       | No fire warning.                                                                                                                                                                                                                |
| FDU failure cleared while the fire burns                   | The fire warning comes at once.                                                                                                                                                                                                 |
| ENG 1 FIRE TEST with Engine 1 fire detection loop A failed | The ENG MASTER 1 FIRE light stays dark; the rest of the test as before.                                                                                                                                                         |
| ENG 2 FIRE TEST with Engine 2 fire detection loop B failed | The ENG MASTER 2 FIRE light stays dark.                                                                                                                                                                                         |
| APU fire on the ground                                     | The external horn variable is set (L:A32NX_FIRE_APU_EXTERNAL_HORN); the horn sound plays at the nose gear.                                                                                                                      |

### Limitations and why

| Limitation                                                            | Why                                                                      | What to do                                         |
|-----------------------------------------------------------------------|--------------------------------------------------------------------------|----------------------------------------------------|
| The APU fire horn is not heard in MSFS 2024.                          | The Asobo horn sound it uses is silent in MSFS 2024 (the MECH call too). | Check the L:var with a SimConnect / variable tool. |
| No APU SHUT OFF pb and no APU FIRE light on the external power panel. | The exterior model has no such panel.                                    | None.                                              |

|                                                |                                                                              |                                   |
|------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------|
| A full MSFS restart is needed after an update. | The systems WASM module, the cockpit behaviour XML and the sound.xml change. | Restart the sim after installing. |
|------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------|

### How to use

1. Open flyPad > Failures > ATA 26 Fire Protection and activate an "... fire detection unit (FDU)" failure.
2. Follow the ECAM (crew awareness); deactivate the failure to restore the unit.

### How to test

| # | Do this                                                                                          | Expected result                                                                                                  |
|---|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| 1 | A32NX at the gate, powered, no failure. Activate <b>Engine 1 fire detection unit (FDU)</b> .     | Amber ENG 1 FIRE DET FAULT, no ENG 1 FIRE LOOP A / B FAULT; STATUS INOP SYS FIRE DET 1.                          |
| 2 | Also activate <b>Engine 1 fire</b> .                                                             | No ENG 1 FIRE warning.                                                                                           |
| 3 | Deactivate <b>Engine 1 fire detection unit (FDU)</b> .                                           | The ENG 1 FIRE warning comes at once. Deactivate the fire.                                                       |
| 4 | Repeat with <b>Engine 2 fire detection unit (FDU)</b> and <b>APU fire detection unit (FDU)</b> . | ENG 2 FIRE DET FAULT / STATUS FIRE DET 2; APU FIRE DET FAULT / STATUS APU FIRE DET.                              |
| 5 | Activate <b>Engine 1 fire detection loop A</b> , then press and hold the ENG 1 FIRE TEST pb.     | The ENG 1 FIRE pushbutton lights as in the normal test, but the FIRE light on the ENG MASTER 1 panel stays dark. |
| 6 | Deactivate it, activate <b>Engine 2 fire detection loop B</b> , hold the ENG 2 FIRE TEST pb.     | The ENG MASTER 2 FIRE light stays dark. With loop A of engine 2 failed instead, it comes on.                     |
| 7 | On the ground with the APU running, activate <b>APU fire</b> .                                   | APU FIRE warning;<br>L:A32NX_FIRE_APU_EXTERNAL_HORN is 1 (the horn itself is silent in MSFS 2024).               |

## 7. Wing tank pump and X FEED valve failures

|                                                                                                                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Aircraft:</b> A32NX   <b>Where:</b> flyPad > Failures > ATA 28 Fuel; overhead FUEL panel; SD FUEL   <b>Status:</b> Tested in the sim by the user: OK (2026-10-04), after the fix that holds the MSFS pump / valve switch. |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### What it is for

Five new failures: **L TK pump 1**, **L TK pump 2**, **R TK pump 1**, **R TK pump 2** and **X FEED valve jammed in place**. The FUEL panel pushbuttons now set a selection, and the systems host moves the MSFS pump or valve switch from it: a failed pump is held off, the jammed valve is held where it was. The FAULT lights, the SD FUEL page and the cautions compare the selection with the real state (A320 FCOM PRO-ABN-FUEL).

- SD FUEL page: amber LO only for a pump that is ON with a low pressure; the X FEED valve is amber when it disagrees with its pushbutton (FCOM colours).
- The FUEL X FEED memo now needs the X FEED pushbutton ON and the valve not fully closed.

### What you should see

| Situation                         | What you should see                                                                                                                                                                      |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| L TK pump 1 failed, pushbutton ON | FAULT light; amber LO on the SD FUEL page (page called). E/WD <b>FUEL L TK PUMP 1 LO PR</b> with -L TK PUMP 1 ..... OFF (shown while the pushbutton is ON). STATUS INOP SYS L TK PUMP 1. |

|                                       |                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Both pumps of one wing failed         | <b>FUEL L TK PUMP 1+2 LO PR</b> (R for the right wing): .IF NO FUEL LEAK: -FUEL X FEED ON (above FL 150 only), -ENG MODE SEL IGN, -L TK PUMP 1 OFF, -L TK PUMP 2 OFF, .WHEN L TK FUEL RQRD: L TK FEED GRVTY ONLY, -FUEL X FEED OFF (below FL 150, X FEED ON), PROC:GRVTY FUEL FEEDING, AVOID NEGATIVE G FACTOR. Each action line goes once done. STATUS INOP SYS L TK PUMPS. |
| X FEED valve jammed, pushbutton moved | The valve stays: amber on the SD FUEL page, <b>FUEL X FEED VALVE FAULT</b> (crew awareness), STATUS INOP SYS FUEL X FEED. With the valve stuck closed, no FUEL X FEED memo.                                                                                                                                                                                                  |

## Limitations and why

| Limitation                                                                            | Why                                                                                                                                                          | What to do                     |
|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| The cautions are inhibited during the takeoff and the landing (phases 3, 4, 5, 7, 8). | The FCOM gives the inhibition as a figure: that of the other single system faults is used.                                                                   | Test at the gate or in cruise. |
| The centre tank has no pump failure.                                                  | The A320neo centre tank has no pumps: its fuel is moved by the jet pumps through the CTR TK L / R XFR valves, whose jams are a feature of their own (below). | None.                          |

## How to use

1. Open flyPad > Failures > ATA 28 Fuel and activate a pump or the X FEED valve failure.
2. Follow the ECAM with the overhead FUEL panel; check the SD FUEL page.

## How to test

| # | Do this                                                                                                                          | Expected result                                                                                                                                                                      |
|---|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | A32NX at the gate, APU or engines running, all fuel pumps ON. Activate <b>L TK pump 1</b> .                                      | After a few seconds: FAULT light in the L TK PUMP 1 pushbutton, SD FUEL page with amber LO, master caution, E/WD FUEL L TK PUMP 1 LO PR with -L TK PUMP 1 ..... OFF.                 |
| 2 | Set the L TK PUMP 1 pushbutton OFF. Open STATUS.                                                                                 | The FAULT light and the OFF line go; no LO on the SD; STATUS INOP SYS L TK PUMP 1.                                                                                                   |
| 3 | Set it ON again and also activate <b>L TK pump 2</b> .                                                                           | FUEL L TK PUMP 1+2 LO PR with its procedure (on the ground, below FL 150: .IF NO FUEL LEAK; -ENG MODE SEL IGN, -L TK PUMP 1 OFF, -L TK PUMP 2 OFF, ...); STATUS INOP SYS L TK PUMPS. |
| 4 | Deactivate both. With the X FEED valve closed, activate <b>X FEED valve jammed in place</b> , then set the X FEED pushbutton ON. | The valve stays closed: amber X FEED valve on the SD, FUEL X FEED VALVE FAULT, STATUS INOP SYS FUEL X FEED, no FUEL X FEED memo.                                                     |
| 5 | Deactivate the failure.                                                                                                          | The valve opens; the FUEL X FEED memo comes; the caution clears.                                                                                                                     |

## 8. Centre tank transfer valve failures and fuel follow-ups

**Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 28 Fuel; overhead FUEL panel; SD FUEL; flyPad checklists | **Status:** Tested in the sim by the user: OK (2026-10-04). Unit tests: FuelPumpValveFaults.spec.ts, FuelPumpsAndValves.spec.ts, FuelPumpValveIndications.spec.ts, CheckItemStates.spec.ts.

## What it is for

Two new failures, **CTR TK L XFR valve jammed in place** and **CTR TK R XFR valve jammed in place**: the centre tank transfer valve of that side stays open or closed, whatever its CTR TK XFR pushbutton commands. The FWC raises the A320 FCOM PRO-ABN-FUEL cautions: FUEL CTR L(R) XFR FAULT (valve not fully closed, or not fully open) and, with both valves stuck the same way, FUEL CTR L + R XFR FAULT, which replaces the two single ones. The SD FUEL page shows the valve amber and boxes the centre tank quantity.

- The centre tank fuel only transfers while an inner tank pump of the same wing runs: the jet pumps are driven by the wing tank pumps (FCOM DSC-28-10). Switching the wing pumps off stops the centre tank transfer.
- flyPad checklist: the securing / parking checklist item **FUEL PUMPS ... OFF** now reads the four wing tank pump pushbuttons, not whether the pumps run (the CTR TK XFR pushbuttons are items of their own in the FCOM).

## What you should see

| Situation                                 | What you should see                                                                                                                                                                                                                                                            |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| L valve stuck open (CTR TK L XFR pb OFF)  | E/WD <b>FUEL CTR L XFR FAULT</b> : -CTR TK L XFR ... OFF, -FUEL X FEED ... ON, -L TK PUMP 1 ... OFF, -L TK PUMP 2 ... OFF, .WHEN CTR TK EMPTY: -L TK PUMP 1 ... ON, -L TK PUMP 2 ... ON, -FUEL X FEED ... OFF (each action line goes once done). STATUS INOP SYS CTR TK L XFR. |
| L valve stuck closed (CTR TK L XFR pb ON) | <b>FUEL CTR L XFR FAULT</b> : -FUEL MODE SEL ... MAN, .IF UNSUCCESSFUL: -FUEL X FEED ... ON, -L TK PUMP 1 / 2 ... OFF, .WHEN CTR TK EMPTY: pumps ON, X FEED OFF. STATUS INOP SYS CTR TK L XFR.                                                                                 |
| Both valves stuck open                    | <b>FUEL CTR L+R XFR FAULT</b> : -CTR TK L XFR ... OFF, -CTR TK R XFR ... OFF; STATUS INOP SYS CTR TK XFR.                                                                                                                                                                      |
| Both valves stuck closed                  | <b>FUEL CTR L+R XFR FAULT</b> : -FUEL MODE SEL ... MAN, .IF UNSUCCESSFUL: CTR AVAIL BY GRAVITY, 2T (4400LBS) UNUSABLE; STATUS: CTR TK USABLE BY GRAVITY, 2T (4400LBS) UNUSABLE, INOP SYS CTR TK XFR.                                                                           |
| SD FUEL page                              | The jammed valve amber; the centre tank quantity boxed.                                                                                                                                                                                                                        |

## Limitations and why

| Limitation                                                                            | Why                                                                                 | What to do                                  |
|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------|
| The cautions are inhibited during the takeoff and the landing (phases 3, 4, 5, 7, 8). | The FCOM gives the inhibition as a figure: that of the other fuel cautions is used. | Test at the gate or in cruise.              |
| No gravity transfer from the centre tank.                                             | Not simulated: with both valves stuck closed, the centre tank fuel stays there.     | Plan with the centre tank fuel as unusable. |

## How to use

1. Open flyPad > Failures > ATA 28 Fuel and activate **CTR TK L XFR valve jammed in place** or **CTR TK R XFR valve jammed in place**.
2. Move the CTR TK XFR pushbutton (or let the automatic control move the valve), then follow the ECAM with the overhead FUEL panel.

## How to test

| # | Do this | Expected result |
|---|---------|-----------------|
|---|---------|-----------------|

|   |                                                                                                                                                                                                                        |                                                                                                                                                                                                 |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | A32NX at the gate with fuel in the centre tank, engines or APU running, all FUEL pushbuttons ON. Activate <b>CTR TK L XFR valve jammed in place</b> while the valve is open, then set the CTR TK L XFR pushbutton OFF. | The valve stays open: master caution, E/WD FUEL CTR L XFR FAULT with -CTR TK L XFR ... OFF done, -FUEL X FEED ... ON, -L TK PUMP 1 ... OFF, -L TK PUMP 2 ... OFF; SD FUEL with the valve amber. |
| 2 | Open the STATUS page.                                                                                                                                                                                                  | INOP SYS: CTR TK L XFR.                                                                                                                                                                         |
| 3 | Also activate <b>CTR TK R XFR valve jammed in place</b> and set the CTR TK R XFR pushbutton OFF.                                                                                                                       | The two single cautions are replaced by FUEL CTR L+R XFR FAULT; STATUS INOP SYS CTR TK XFR.                                                                                                     |
| 4 | Deactivate both failures, CTR TK XFR pushbuttons ON. Set L TK PUMP 1 and 2 OFF.                                                                                                                                        | The left centre tank transfer stops (no jet pump drive) while the right one goes on.                                                                                                            |
| 5 | Set every wing tank pump pushbutton OFF and open the flyPad checklist with the FUEL PUMPS ... OFF item.                                                                                                                | The item is checked from the four wing tank pump pushbuttons.                                                                                                                                   |

## 9. Engine flameout and seizure failures, in-flight relight

**Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 72 Engine; ENG MASTER 1 / 2, ENG MODE selector; E/WD, STATUS, SD ENG | **Status:** Tested in the sim by the user on develop: OK, tests T1-T9 (2026-10-05); the LEAP-1A relight envelope (a27ba6c35): sim-tested OK 2026-10-05 (recorded), tests L1-L2. Rust tests (engine\_failure.rs of systems and a320\_systems, incl. the LEAP-1A envelope corners and two test-bed relights; full-aircraft tests incl. the ENG FIRE pb LP valve starvation), unit tests: EngineFailAlerts.spec.ts, A32NX\_FADEC.spec.ts, Eng.spec.ts, a320.spec.ts. The end of an in-flight start without N2 overshoot: sim-tested OK on develop 2026-10-06 (relight test T1, recorded), native test engine\_start\_throttle\_hold\_test. The engine fuel valve hold: not sim-tested on the A32NX (Rust tests sim\_fuel\_valve.rs). The stall during a relight fix: not sim-tested (installed 2026-10-08, no visible effect expected; native test T8). Not sim-tested on this branch.

### What it is for

Four new failures in a new flyPad chapter, ATA 72 Engine. A **flameout** cuts the engine fuel once: the crew can relight the engine with the ENG MASTER OFF then ON, inside the in-flight relight envelope. A **seizure** also stops the core: no relight while the failure is active. The FWC gives the A320 FCOM ENG 1(2) FAIL and ENG 1(2) SHUT DOWN alerts with their STATUS, and the flyPad failures page now describes the ATA 28 Fuel and ATA 72 Engine chapters like the others.

| Id    | flyPad failure (ATA 72 Engine)            | What the code does                                                                                                                                                                                                                                                                |
|-------|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 72000 | Engine 1 flameout (crew relight possible) | Cuts the engine 1 fuel (latched). The engine runs down and windmills; the FADEC does not relight it. ENG MASTER 1 OFF then ON inside the relight envelope relights it; it then runs normally while the failure stays listed. Deactivate and activate it again for a new flameout. |
| 72001 | Engine 2 flameout (crew relight possible) | The same for engine 2.                                                                                                                                                                                                                                                            |
| 72010 | Engine 1 seizure (no relight)             | Cuts the fuel; the core (N2) stops in about 3 s while N1 windmills. No relight while the failure is active; once it is cleared the engine relights by the normal procedure.                                                                                                       |
| 72011 | Engine 2 seizure (no relight)             | The same for engine 2.                                                                                                                                                                                                                                                            |

The **relight envelope** comes from the LEAP-1A chart ENG RELIGHT IN FLIGHT of the Air Arabia A320 QRH (18-Aug-21, ABN-19.06A, A6-ATA to ATF, CFM LEAP-1A engines, the engines FBW models). It replaces the PW1100G chart of the IndiGo A320neo FCOM (PDF page 4761), used until 2026-10-05 while no LEAP-1A chart was available. Vc is the indicated airspeed, Zp the pressure altitude.

| Zone | Where | Needs |
|------|-------|-------|
|------|-------|-------|

|                                           |                                                                                               |                                                                                                                            |
|-------------------------------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| 3, stabilized windmill relight (N2 > 8 %) | 270-350 kt up to 27 000 ft; the ceiling falls from 27 000 ft at 330 kt to 24 000 ft at 350 kt | ENG MODE IGN, ENG MASTER OFF then ON                                                                                       |
| 1, starter assisted relight               | 135-270 kt up to 15 000 ft; 200-270 kt from 15 000 to 27 000 ft                               | ENG MODE IGN, ENG MASTER OFF then ON, and starter air (APU BLEED, or X BLEED OPEN with the other engine)                   |
| 2, windmill quick relight (t < 20 s)      | 230-270 kt up to 20 000 ft                                                                    | ENG MASTER OFF then ON within 20 s, with N2 above 15 % (design choice, not on the LEAP-1A chart), at any ENG MODE position |

- Above these limits (e.g. in cruise at FL 350) no relight lights up.
- A windmilling engine (no combustion, in flight) keeps  $N2 = 8 \% \times IAS / 270 \text{ kt}$  and  $N1 = 1.5 \times N2$  (at 300 kt: N2 8.9 %, N1 13.3 %); on the ground it runs down to 0. Its fuel flow shows 0 and its oil pressure falls with N2.
- During an in-flight relight both igniters are supplied (A320 FCOM DSC-70-80-30), whatever the ENG MODE selector: the SD ENG page shows A B.
- The SD ENG page reads the N2 the FADEC shows, so a seized core reads 0 there as on the E/WD.
- ENG ALL ENGINES FAILURE gets its FCOM flight phase inhibition (phases 1-4 and 8-10). Its OPT RELIGHT SPD line reads 270/.77, the LEAP-1A value of the same QRH (ABN-19.01A ALL ENG FAIL; the CFM56 aircraft use 300/.77); the removal of the unused 280/300/260 lines is on core/fcom-audit-fixes.
- **End of an in-flight start (fix):** an engine started or relit in flight used to overshoot to 110-114 % N2 for 3-7 s at the end of the start, because the thrust loop wound the MSFS throttle up while the engine was out. In flight, the MSFS throttle is now held at idle while the engine is out or starting, until the engine runs at idle; then the engine accelerates to the lever or A/THR demand (A320 FCOM DSC-70-80-40, QRH ENG RELIGHT IN FLIGHT "When idle reached"). Ground starts are unchanged, and reverse thrust passes through this hold (a first version of it blocked the reverse thrust).
- **Stall during a relight (fix):** the hold ends when the commanded N1 reaches idle. With the "Engine 1(2) compressor stall" failure active at the end of a relight, the stall N1 loss (up to 15 %) was counted in the commanded N1 and the hold ended about 0.8 s early; the stall N1 loss is now taken out of that check (native test T8: a relight with a stall).
- **ENG MODE selector (fix):** the FADEC read a selector variable that does not exist (always NORM), so it never saw IGN/START; it now reads the real selector.
- **Engine fuel valve hold (fix):** MSFS can reopen the engine fuel valve (valves 13 and 14 of the A32NX engine fuel cut) on its own after the systems closed it. The systems now command the valve again whenever MSFS disagrees, and once when a flight is loaded.

## What you should see

| Situation          | What you should see                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Flameout in flight | N2, EGT and FF run down, FF 0, N1 / N2 settle at the windmill values. Amber <b>ENG 1 FAIL</b> , single chime, MASTER CAUT, SD ENG page: -ENG MODE SEL ..... IGN, -THR LEVER 1 ..... IDLE, .IF NO RELIGHT AFT 30S;; 30 s after the alert -ENG MASTER 1 ..... OFF, .IF DAMAGE: -ENG 1 FIRE P/B ..... PUSH, -AGENT1 AFTER 10S ..... DISCH, .IF NO DAMAGE: -ENG 1 RELIGHT ..... CONSIDER (A320 FCOM PRO-ABN-ENG ENG 1(2) FAIL). LAND ASAP amber. The ENG 1 FAULT light stays off (FCOM: a flameout does not light it). |

|                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ENG MASTER 1 OFF in flight (after a failure or not)</b> | ENG 1 FAIL goes, <b>ENG 1 SHUT DOWN</b> comes: .IF NO FUEL LEAK; - IMBALANCE ..... MONITOR, -TCAS MODE SEL ..... TA (with the wing anti-ice on: the pack of the affected side OFF and X BLEED OPEN; with the ENG FIRE pb pushed: X BLEED SHUT, WING ANTI ICE OFF, AVOID ICING CONDITIONS). STATUS: IF PERF PERMITS: X BLEED ..... OPEN, IF NO ENG 1 DAMAGE: CONSIDER ENG 1 RELIGHT, CAT 3 SINGLE ONLY, ONE PACK ONLY IF WAI ON; INOP SYS: CAT 3 DUAL, ENG 1 BLEED, PACK 1, MAIN GALLEY, GEN 1, G ENG 1 PUMP (WING A. ICE with the ENG FIRE pb pushed). |
| <b>Relight inside the envelope</b>                         | Light-up: FF, N2 and EGT rise, A B igniters on the SD ENG page, AVAIL; the engine alerts and LAND ASAP clear.                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Relight outside the envelope</b>                        | No light-up: N2 stays at the windmill value, FF 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Seizure</b>                                             | N2 to 0 in about 3 s, N1 windmills, oil pressure 0 (red), ENG 2 FAIL as above; GEN 2 and the yellow engine pump lost.                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Both engines failed</b>                                 | Red ENG ALL ENGINES FAILURE; the single ENG 1 / ENG 2 FAIL alerts do not show while it is active. Relighting one engine brings the single alert of the other one.                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>On the ground</b>                                       | ENG 1 FAIL with the ground lines (-THR LEVER 1 ..... IDLE, -ENG MASTER 1 ..... OFF, .IF DAMAGE: ..., -AGENT 1 ..... DISCH, .IF NO DAMAGE: ...), no LAND ASAP; N1 / N2 run down to 0. ENG MASTER OFF gives no ENG SHUT DOWN in flight phases 1-2 and 9-10 (FCOM: phases 3-8 only).                                                                                                                                                                                                                                                                      |

## Limitations and design choices

| Limitation                                                                                               | Why                                                                                                                                                 | What to do                                                                  |
|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| The relight envelope applies to every in-flight relight, also after a crew shutdown without any failure. | Design choice: MSFS would otherwise relight anywhere; the QRH chart is the limit of any relight.                                                    | Relight inside the envelope (descend and adjust the speed).                 |
| The envelope corners are read off the QRH chart grid.                                                    | The chart gives no table; the calibrated airspeed is taken as the indicated airspeed.                                                               | -                                                                           |
| The quick relight also needs N2 above 15 %.                                                              | Design choice kept from the PW1100G chart (the LEAP-1A chart only says $t < 20$ s): the sign of an engine that was running before the MASTER cycle. | An engine that has windmilled for a while needs ENG MODE IGN (zone 1 or 3). |
| The windmill N1 / N2 values are design values.                                                           | The chart only marks zone 3 "N2 > 8 %"; N2 is taken proportional to the airspeed, 8 % at 270 kt, and $N1 = 1.5 \times N2$ .                         | -                                                                           |
| The seizure stops the core in about 3 s.                                                                 | Design choice: the FCOM gives no rate.                                                                                                              | -                                                                           |
| ENG 1(2) FAIL does not come back after the ENG MASTER OFF, even if the relight does not light up.        | Design choice: the MASTER OFF disarms the alert; ENG SHUT DOWN is shown meanwhile.                                                                  | After a failed attempt (no light-up within 30 s), MASTER OFF and try again. |
| While ENG ALL ENGINES FAILURE is active, the single engine alerts are not shown.                         | Design choice: its procedure replaces them.                                                                                                         | -                                                                           |

|                                                                                                    |                                                                                                                                                                         |                                   |
|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| In flight, the HP fuel valve opening at N2 > 15 % is not modelled.                                 | The fuel comes back when the engine lights up inside the envelope (on the ground the start sequence of "Engine ignition and start failures" opens the fuel at 22 % N2). | -                                 |
| The idle hold after an in-flight start ends after a fixed time if MSFS settles below the FBW idle. | Design choice (no FCOM value): MSFS can end its start slightly below the FBW idle core speed.                                                                           | -                                 |
| A full MSFS restart is needed after an update.                                                     | The systems WASM module, the FADEC WASM module and the fbw WASM module (throttle hold) change.                                                                          | Restart the sim after installing. |

## How to use

1. Open flyPad > Failures > ATA 72 Engine and activate a flameout or a seizure.
2. Follow the ECAM (ENG FAIL, then ENG SHUT DOWN). To relight, get inside the envelope (table above), ENG MODE IGN, ENG MASTER OFF for 30 s, then ON. For a quick relight in zone 2, set the MASTER OFF then ON at once (within 20 s, N2 above 15 %, 230-270 kt up to 20 000 ft).
3. Clear the failure in the flyPad after the test; a seizure must be cleared before a relight.

## How to test

From the test plan [reports/test-plan-engine-failures-a1.md](#). Setups: S1 cruise 60 t, FL 350, M.78; S2 60 t, 10 000 ft, 300 kt; S3 60 t, 15 000 ft, 210 kt, APU running with APU BLEED ON; S4 on the ground, both engines at idle.

| # | Do this                                                                                                           | Expected result                                                                                                                                                                  |
|---|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | T1 (S1): activate <b>Engine 1 flameout (crew relight possible)</b> .                                              | ENG 1 FAIL with its lines, LAND ASAP, windmill N1 / N2, FF 0, oil pressure falling with N2; the 30 s lines come after 30 s. ENG MODE IGN and THR LEVER 1 IDLE clear their lines. |
| 2 | T1: after 30 s, ENG MASTER 1 OFF; then MASTER ON at FL 350.                                                       | ENG 1 SHUT DOWN with its STATUS and INOP SYS; at FL 350 no light-up (outside the envelope), A B igniters on the SD ENG page.                                                     |
| 3 | T2: descend to 20 000 ft or below at 280-300 kt; ENG MODE IGN, MASTER 1 OFF 30 s, ON.                             | Light-up at once; the engine back at idle in about 30-60 s; all engine 1 alerts and LAND ASAP clear.                                                                             |
| 4 | T3 (S3): flameout; MASTER 1 OFF; APU BLEED OFF, X BLEED not open; 30 s; MASTER 1 ON.                              | No light-up (210 kt is below zone 3, no starter air). MASTER OFF, APU BLEED ON, 30 s, MASTER ON: the start valve opens and the engine lights up.                                 |
| 5 | T4 (S2 slowed to 245 kt below 13 000 ft, ENG MODE NORM): flameout, then at once MASTER 1 OFF then ON within 20 s. | Quick relight at NORM; the ENG MODE knob stays at NORM and still works afterwards. Waiting until N2 is below 15 % before the MASTER OFF / ON: no light-up.                       |
| 6 | T5 (S2): activate <b>Engine 2 seizure (no relight)</b> .                                                          | N2 to 0 in about 3 s, N1 windmills, oil pressure 0, ENG 2 FAIL; MASTER 2 OFF 30 s / ON at 300 kt: no light-up. Clear the seizure, MASTER 2 OFF 30 s / ON: light-up.              |
| 7 | T6 (S2, optional): activate both flameouts.                                                                       | ENG ALL ENGINES FAILURE without the single ENG FAIL alerts; relighting engine 1 brings ENG 2 FAIL.                                                                               |
| 8 | T7 (S1): no failure, MASTER 1 OFF at FL 350, then ON with ENG MODE IGN.                                           | ENG 1 SHUT DOWN; no light-up at FL 350; at FL 200 / 290 kt, MASTER OFF 30 s then ON: light-up.                                                                                   |

|    |                                                                                                                                               |                                                                                                                                                                                                           |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9  | T8 (S4): activate <b>Engine 1 flameout (crew relight possible)</b> on the ground.                                                             | ENG 1 FAIL with the ground lines, no LAND ASAP, N1 / N2 to 0; MASTER 1 OFF gives no ENG 1 SHUT DOWN; a normal start works.                                                                                |
| 10 | T9: normal flight (starts, single-engine taxi, takeoff to descent, shutdown); on the ground at idle, push the ENG 1 FIRE pb with MASTER 1 ON. | No ENG FAIL / SHUT DOWN alert in the normal flight. With the FIRE pb: ENG 1 SHUT DOWN at once, no ENG 1 FAIL; the engine stops after about 100 s on its residual fuel (LP valve starvation of core/fire). |

LEAP-1A envelope re-test ( `reports/test-card-a32nx-leap-relight.md` , recorded over SimConnect, OK 2026-10-05): two points the old PW1100G envelope refused. L1 setup: FL 150, 250 kt, AP and A/THR ON, APU OFF, ENG MODE NORM, both engines running. L2 setup: climbed to FL 265 at 290 kt (279 kt at the relight), AP and A/THR ON, APU OFF, X BLEED AUTO, ENG MODE NORM.

| # | Do this                                                                                                                                                                                                                                           | Expected result                                                                                                                                                                  |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | L1 (quick relight at FL 150): ENG MASTER 1 OFF, count 5 s, ENG MASTER 1 ON.                                                                                                                                                                       | Engine 1 fuel flow comes back within a few seconds, N2 and EGT rise, engine 1 back at idle; no ENG 1 FAIL stays; the ENG MODE selector stays at NORM.                            |
| 2 | L2 (windmill relight at FL 265): activate <b>Engine 1 flameout (crew relight possible)</b> ; THR LEVER 1 IDLE; ENG MODE IGN/START; ENG MASTER 1 OFF, count 10 s; ENG MASTER 1 ON. At idle: ENG MODE NORM, THR LEVER 1 CL, deactivate the failure. | Engine 1 fuel flow appears within a few seconds, N2 and EGT rise, engine 1 at idle within about 60 s, ENG 1 FAIL goes away (the aircraft may sink a few hundred feet meanwhile). |

End of the in-flight start ( `reports/test-card-relight-batch-2026-10-06.md` T1, recorded, OK 2026-10-06 on develop). Setup: FL 150, 250 kt, AP and A/THR ON, APU OFF, ENG MODE NORM, both engines running, both thrust levers in CL.

| # | Do this                                                                                               | Expected result                                                                                                                                                                                                                                                 |
|---|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | ENG MASTER 1 OFF, count 5 s, ENG MASTER 1 ON. Leave both levers in CL and watch engine 1 N2 for 40 s. | Engine 1 relights within about 10 s; its N2 rises to idle (about 70 %) and holds there until the engine runs, then rises in about 5 s to the N2 of engine 2 (about 93 %). Engine 1 N2 never goes more than 1-2 % above engine 2 N2 (before the fix: 110-114 %). |

## 10. Engine oil failures: leak, filter clog, overheat

**Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 79 Oil; SD ENGINE and CRUISE pages; E/WD | **Status:** Sim-tested by the user on develop, data-checked OK (2026-10-07 retest card T1: no false oil alert in a climb to FL 300; the three oil failures together in cruise: overheat 81 to 158 °C with ENG 2 OIL HI TEMP, leak to the red oil pressure below 13 PSI with ENG 1 OIL LO PR, then the shutdown). The later tuning (gradual cool-down after the overheat, overheat times): not sim-tested on the A32NX. Rust tests `oil_failure.rs`, native tests `engine_oil_failures_test` and `oil_system_test`, unit tests `EngineOilAlerts.spec.ts`, `EngOilIndications.spec.ts`.

### What it is for

Six new failures: **Engine 1 oil leak**, **Engine 2 oil leak**, **Engine 1 oil filter clog**, **Engine 2 oil filter clog**, **Engine 1 oil overheat** and **Engine 2 oil overheat**, with the oil indications and the ECAM alerts of the A320 FCOM (DSC-70-90-40, PRO-ABN-ENG).

The FADEC computes the oil quantity, pressure and temperature again: its oil model had been dead since a FADEC migration and is restored (from the A380X FADEC, with the A32NX engine imbalance effect on the oil pressure). The failures act on that model:

- **Leak:** the oil system empties at 2 qt per minute of simulation time (the SD tank reading falls about 1.6 qt/min). The pump keeps its normal pressure down to 2.5 qt in the whole system and gives none at 0.6 qt, so the pressure

falls once the tank reads nearly empty.

- **Filter clog:** CLOG on the SD ENGINE page and the ENG OIL FILTER CLOG caution, no other effect.
- **Overheat:** the oil goes to 135 °C at idle and 175 °C at 100 % N2 (about 165 °C at cruise), with a 60 s time constant: ENG OIL HI TEMP after about 2 minutes in cruise; THR LEVER IDLE brings it below 155 °C within about 30 s. Once the failure is cleared the oil cools down gradually to its normal temperature.
- **SD ENGINE page:** the FCOM 0-22 QT scale; the quantity pulses below 3.25 QT (until 4.75 QT); the oil pressure red below 13 PSI, pulsing between 13 and 16 PSI with N2 above 75 % (until it is back above 20 PSI); amber CLOG under the oil pressure. The CRUISE page shows the same FADEC oil quantity.
- **Fixes found in the sim tests:** the SD oil quantity read nearly empty in every climb (the oil gulping formula got the thrust in newtons instead of pounds), and the pressure followed the tank reading, which gave a false ENG OIL LO PR in a normal climb: the pump pressure now follows the oil of the whole system. The overheat stopped far below its target because MSFS pulls the oil temperature back to its own model: the FADEC now keeps its own overheat temperature.
- The start N2 and fuel flow of the FADEC model also blend smoothly at low N2 now (they overshoot below about 10 % N2).

## What you should see

| Situation                       | What you should see                                                                                                                                                                                                              |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Oil leak, after several minutes | Oil quantity falling, then pulsing below 3.25 QT; the oil pressure falls and turns red below 13 PSI; red <b>ENG 1 OIL LO PR</b> , MASTER WARN and continuous chime: -THR LEVER 1 ... IDLE, -ENG MASTER 1 ... OFF.                |
| Oil filter clog                 | Amber <b>ENG 2 OIL FILTER CLOG</b> (title only, single chime) and amber CLOG on the SD ENGINE page.                                                                                                                              |
| Oil overheat                    | The oil temperature rises from about 85 °C, pulses above 140 °C and is amber above 155 °C; amber <b>ENG 2 OIL HI TEMP</b> : -THR LEVER 2 ... IDLE, -ENG MASTER 2 ... OFF. It also comes after more than 15 minutes above 140 °C. |
| Lever at idle                   | The -THR LEVER line goes.                                                                                                                                                                                                        |
| ENG MASTER OFF                  | The oil alert ends and ENG SHUT DOWN takes over (FCOM associated procedure).                                                                                                                                                     |

## Limitations and why

| Limitation                                                                                                     | Why                                                                                                  | What to do      |
|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-----------------|
| Inhibitions: OIL LO PR in flight phases 1 and 10, OIL HI TEMP in 4, 5, 7, 8, OIL FILTER CLOG in 3, 4, 5, 7, 8. | The FCOM figures (2019 FCOM pages 2281-2283).                                                        | Test in cruise. |
| The oil alerts and CLOG only show with the engine running and its ENG MASTER ON.                               | Design choice: a stopped engine has no oil pressure, and the procedures end with the ENG MASTER OFF. | -               |
| The leak rate, the pressure loss and the overheat temperatures are design values.                              | The FCOM gives no rates; the overheat cause is not modelled either.                                  | -               |
| The leak rate is per simulation minute.                                                                        | At 4x sim rate the oil goes four times faster on the real clock.                                     | -               |

|                                                                                  |                                                                                                                  |                                   |
|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| The upper oil pressure scale stays the FBW one (130 PSI, advisory from 129 PSI). | The FCOM 0-100 PSI scale is the CFM56 one; the FBW LEAP-1A oil pressure reaches about 90 PSI at take-off thrust. | -                                 |
| A full MSFS restart is needed after an update.                                   | The systems and FADEC WASM modules change.                                                                       | Restart the sim after installing. |

### How to use

1. Call the SD ENGINE page (ECAM control panel ENG key).
2. Open flyPad > Failures > ATA 79 Oil and activate a leak, a filter clog or an overheat.
3. Follow the ECAM (THR LEVER IDLE, then ENG MASTER OFF if needed). Deactivate the failure after the test.

### How to test

From `reports/test-card-retest-2026-10-07.md` T1 (recorded OK on develop 2026-10-07). Setup: normal departure, takeoff and climb at CL with A/THR ON, then cruise at FL 300-390.

| # | Do this                                                                                                                           | Expected result                                                                                                                                                                                    |
|---|-----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Climb normally and watch the oil indications.                                                                                     | No OIL LO PR, no pulsing oil quantity; oil quantity about 10-15 qt in the climb.                                                                                                                   |
| 2 | In cruise, SD ENGINE page. Activate <b>Engine 2 oil filter clog</b> , <b>Engine 2 oil overheat</b> and <b>Engine 1 oil leak</b> . | Amber ENG 2 OIL FILTER CLOG and CLOG on the SD; engine 2 oil temperature rises past 155 °C within about 3 minutes, then amber ENG 2 OIL HI TEMP with -THR LEVER 2 ... IDLE, -ENG MASTER 2 ... OFF. |
| 3 | Deactivate <b>Engine 2 oil filter clog</b> and <b>Engine 2 oil overheat</b> .                                                     | Both engine 2 alerts and CLOG go; the temperature falls gradually.                                                                                                                                 |
| 4 | Wait for the leak (about 6 to 10 minutes after it began).                                                                         | Engine 1 oil quantity falls about 2 QT/min and pulses below 3.25 QT; the pressure falls and turns red below 13 PSI; red ENG 1 OIL LO PR with MASTER WARN.                                          |
| 5 | THR LEVER 1 IDLE, ENG MASTER 1 OFF. Deactivate <b>Engine 1 oil leak</b> .                                                         | The lever line goes; after the MASTER OFF, ENG 1 SHUT DOWN replaces the oil alert.                                                                                                                 |

## 11. Engine fuel filter clog

**Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 73 Engine - Fuel And Control; SD ENGINE page; E/WD | **Status:** Not sim-tested (offline: Rust `fuel_filter_failure.rs` tests, `EngFuelFilterIndication.spec`, `EngineFuelFilterAlerts.spec`, failure list specs). The A380X version of the same code path was sim-tested on develop (2026-10-08).

### What it is for

Two new failures, **Engine 1 fuel filter clog** and **Engine 2 fuel filter clog**: "the pressure loss across the fuel filter is excessive" (FCOM DSC-70-90-40). Amber **CLOG** below FUEL USED on the SD ENGINE page while the engine runs, and the **ENG 1(2) FUEL FILTER CLOG** caution (crew awareness, "short term engine operation is not affected"). No effect on the engine: the filter is bypassed.

### Limitations and why

| Limitation                                   | Why                      | What to do                       |
|----------------------------------------------|--------------------------|----------------------------------|
| Inhibited in flight phases 3, 4, 5, 7 and 8. | The FCOM procedure page. | Test on the ground or in cruise. |

|                                       |                                                   |   |
|---------------------------------------|---------------------------------------------------|---|
| No fuel system contamination failure. | Not modelled; the clog only gives the indication. | - |
|---------------------------------------|---------------------------------------------------|---|

## How to use

1. Call the SD ENGINE page (ECAM control panel ENG key).
2. Open flyPad > Failures > ATA 73 and activate **Engine 1 fuel filter clog** with engine 1 running.
3. Deactivate it after the test.

## How to test

On the ground with engine 1 running at idle (flight phase 1 or 2), SD ENGINE page.

| # | Do this                                     | Expected result                                                                                                         |
|---|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| 1 | Activate <b>Engine 1 fuel filter clog</b> . | Amber CLOG below FUEL USED of engine 1; amber ENG 1 FUEL FILTER CLOG with MASTER CAUT. Engine 1 keeps running normally. |
| 2 | Deactivate it.                              | CLOG and the caution go.                                                                                                |

## 12. Engine ignition and start failures, FCOM start sequence

**Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 74 Ignition, ATA 80 Starting; ENG MODE, ENG MASTER, ENG MAN START; E/WD, STATUS, SD ENGINE | **Status:** Sim-tested by the user on develop: OK (2026-10-06 integration flight S7, recorded: a ground start with Engine 1 igniter A failed, the automatic dry crank and the new start with both igniters; the dry crank with ENG MAN START). The other start failures: not sim-tested (Rust tests). Rust tests engine\_start.rs, native test start\_sequence\_test, unit tests EngineStartAlerts.spec.ts, StartIndications.spec.ts.

## What it is for

The FBW FADEC ran the engine model but not what a real FADEC does during a start. The systems now run the start sequence of the A320 FCOM (DSC-70-80-30, DSC-70-80-40):

- **Igniters:** one igniter for a first automatic start on the ground, alternated between starts; both for a new attempt, a manual start and an in-flight start. Continuous ignition with the ENG MODE selector at IGN/START (one igniter, both if it has failed), with the green IGNITION memo. The SD igniter indication now comes from this sequence.
- **Automatic start:** ignition from 16 % N2, fuel (HP valve) at 22 % N2, ignition off and start valve closed at 50 % N2. On the ground, a start that does not light up within 18 s, stalls, hangs or goes above 725 °C EGT is aborted, dry cranked for 30 s and tried again with both igniters, up to three attempts; then the ENG FAULT light (ENG MASTER panel) until ENG MASTER OFF.
- **ENG MAN START pushbuttons:** manual start, dry crank (ENG MODE CRANK, MAN START ON, MASTER OFF) and wet crank. The MSFS starter turns the core while the sequence keeps the fuel cut, with no EGT rise.
- The SD ENGINE page shows the start indications at CRANK too.
- Tuning: an engine that is OFF only shows running once MSFS really burns fuel in it.

| flyPad failure (both engines)                     | ATA         | What the code does                                                                                  |
|---------------------------------------------------|-------------|-----------------------------------------------------------------------------------------------------|
| Engine 1(2) igniter A / igniter B                 | 74 Ignition | The igniter does not light the engine: ENG IGN A (B) FAULT; a start that uses it does not light up. |
| Engine 1(2) start valve stuck closed / stuck open | 80 Starting | The start valve stays as it is whatever the FADEC commands: ENG START VALVE FAULT.                  |
| Engine 1(2) hot start (start EGT over limit)      | 80 Starting | The start EGT overshoots: START FAULT EGT OVERLIMIT, abort and new attempts.                        |

|                                              |                |                                                                                                                            |
|----------------------------------------------|----------------|----------------------------------------------------------------------------------------------------------------------------|
| Engine 1(2) hung start (N2 stops below idle) | 80<br>Starting | N2 hangs below idle: START FAULT HUNG START.                                                                               |
| Engine 1(2) stall during the start           | 80<br>Starting | N2 hangs and the EGT overshoots: START FAULT STALL.                                                                        |
| Engine 1(2) starter shaft shear              | 80<br>Starting | The starter gets air but does not turn the engine: no start, no starter assisted relight; START FAULT STARTER SHAFT SHEAR. |

## What you should see

| Situation                                   | What you should see                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Start fault on the ground (automatic start) | Amber <b>ENG 1 START FAULT</b> with the cause (IGNITION FAULT, STALL, EGT OVERLIMIT, HUNG START, STARTER SHAFT SHEAR, LO START AIR PRESS or THR LEVER NOT AT IDLE), then AUTO CRANK IN PROGRESS during the dry crank, NEW START IN PROGRESS during the next attempt, and -ENG MASTER 1 ... OFF once the start is aborted. |
| Start fault, manual start or in flight      | Manual start on the ground: -ENG MASTER 1 ... OFF, -MAN START 1 ... OFF. In flight: -ENG MASTER 1 ... OFF. Starter shaft shear in flight: WINDMILL START ONLY, and STATUS ENG 1 WINDMILL START ONLY. Low start air: -BLEED AIR SUPPLY ... CHECK; lever not at idle: -THR LEVER 1 ... IDLE.                                |
| Start valve stuck open after the start      | <b>ENG 1 START VALVE FAULT</b> , START VALVE NOT CLOSED: -APU BLEED ... OFF (engine 1), -X BLEED ... SHUT; in flight -ENG 1 BLEED ... OFF, -MAN START 1 ... OFF, -WING ANTI ICE ... OFF, AVOID ICING CONDITIONS (with its STATUS); on the ground -ENG MASTER 1 ... OFF.                                                   |
| Start valve stuck closed at a start         | ENG 1 START VALVE FAULT, START VALVE NOT OPEN: -X BLEED ... OPEN (other engine running), -APU BLEED ... ON (APU available, below FL 200), .IF UNSUCCESSFUL: MAN START / ENG MASTER OFF.                                                                                                                                   |
| Igniter failed                              | <b>ENG 1 IGN A FAULT</b> (IGN B FAULT) crew awareness, STATUS INOP SYS ENG 1 IGN A (B); both failed: ENG 1 IGN A+B FAULT, AVOID ADVERSE CONDITIONS, INOP SYS ENG 1 IGN.                                                                                                                                                   |
| ENG MODE IGN/START with the engines running | Green IGNITION memo.                                                                                                                                                                                                                                                                                                      |

## Limitations and why

| Limitation                                                                                                                                           | Why                                                         | What to do                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|----------------------------------|
| The numbers are those of the CFM56-5B FCOM.                                                                                                          | FBW models the LEAP-1A, for which no FCOM is available.     | -                                |
| The 30 s automatic dry crank is a design value.                                                                                                      | The FCOM gives no length (30 s is its minimum ventilation). | -                                |
| No starter time limit.                                                                                                                               | The A320 FCOM gives none.                                   | -                                |
| The SD still shows the selected igniter when that igniter has failed.                                                                                | Design choice: the FCOM shows the igniter energized.        | -                                |
| Inhibitions: the three cautions in flight phases 3, 4, 5, 7, 8; IGN A or B FAULT also in 6; START FAULT also in 6 when due to the lever not at idle. | The FCOM figures.                                           | Test on the ground or in cruise. |

|                                                                 |                                                                                            |                                                                              |
|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| The "EEC control of start valve failed" lines are not modelled. | They are for IAE and PW engines.                                                           | -                                                                            |
| An aborted automatic start takes about 2.5 minutes in the sim.  | Up to three attempts with 30 s dry cranks.                                                 | Use the igniter test below; the other abort paths are covered by Rust tests. |
| A full MSFS restart is needed after an update.                  | The systems and FADEC WASM modules and the cockpit behaviour XML (ENG FAULT light) change. | Restart the sim after installing.                                            |

## How to use

1. Open flyPad > Failures > ATA 74 Ignition or ATA 80 Starting and activate a failure before the start.
2. Start the engine as usual (ENG MODE IGN/START, ENG MASTER ON) and follow the ECAM; the FADEC aborts and retries a failed automatic start on the ground by itself.
3. For a dry crank: ENG MODE CRANK, ENG MAN START ON with the ENG MASTER OFF.

## How to test

From the integration plan `reports/test-plan-integration-2026-10-06.md` S7 (recorded OK on develop 2026-10-06).  
 Setup: A32NX cold and dark at the gate, freshly loaded, external power or batteries ON, APU AVAIL, APU BLEED ON, parking brake ON, thrust levers IDLE.

| # | Do this                                                                                                                                               | Expected result                                                                                                                                                                                              |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | flyPad > Failures > ATA 74 Ignition: activate <b>Engine 1 igniter A</b> . ENG MODE IGN/START, ENG MASTER 1 ON. Wait 75 s.                             | About 20 s after MASTER ON: amber ENG 1 START FAULT, IGNITION FAULT, AUTO CRANK IN PROGRESS and amber ENG 1 IGN A FAULT; SD ENG: N2 about 20-30 %, no EGT rise, start valve open, igniter A shown then gone. |
| 2 | Keep waiting.                                                                                                                                         | About 50 s after MASTER ON: NEW START IN PROGRESS, SD igniters A B, EGT rises, engine 1 starts to idle, START FAULT goes; the ENG 1 FAULT light stays off. Deactivate the failure.                           |
| 3 | ENG MODE CRANK; ENG MAN START 2 pb ON (overhead, guarded) for 20 s, then OFF; ENG MODE NORM.                                                          | SD ENG: engine 2 start valve open, N2 about 20-25 % and steady, no fuel flow, no EGT rise, no igniter; engine 2 does not start. Then the valve closes and N2 runs down. No ENG 2 START FAULT.                |
| 4 | Optional (not sim-tested): activate <b>Engine 2 starter shaft shear</b> , ENG MODE IGN/START, ENG MASTER 2 ON.                                        | N2 does not rise; ENG 2 START FAULT with STARTER SHAFT SHEAR, -ENG MASTER 2 ... OFF, -MAN START 2 ... OFF. ENG MASTER 2 OFF and deactivate the failure.                                                      |
| 5 | Optional (not sim-tested): engine 1 off, activate <b>Engine 1 start valve stuck open</b> , then start engine 1 (ENG MODE IGN/START, ENG MASTER 1 ON). | Above 50 % N2 the valve stays open: ENG 1 START VALVE FAULT, START VALVE NOT CLOSED with its ground lines (-APU BLEED ... OFF, -X BLEED ... SHUT, -ENG MASTER 1 ... OFF).                                    |

## 13. Engine stall, EGT overtemperature, overspeed and vibration failures

**Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 72 Engine, ATA 77 Engine Indicating; E/WD, SD ENGINE | **Status:** Sim-tested by the user on develop: OK (2026-10-06 integration flight, recorded: S8 T3 stall in the climb, S9 T1 EGT overtemperature with high vibration). VIB N2 then stayed at 4.6 at idle, above the 4.3 advisory: the vibration tuning that brings it below at idle is not sim-tested. The overspeed indication: not sim-tested. Rust tests engine\_malfunction.rs, unit tests EngineParameterAlerts.spec.ts, EngineLimits.spec.ts, EngVibration.spec.ts, Eng.spec.ts.

## What it is for

Eight new failures; the engine keeps running:

| flyPad failure (both engines)                | ATA                  | What the code does                                                                                                                                                                                                                                |
|----------------------------------------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Engine 1(2) compressor stall (above 60 % N1) | 72 Engine            | Above 60 % N1 command the engine stalls: it loses N1 (about 15 %), the EGT rises (about 100 °C or more), N1 and N2 fluctuate, N1 vibrates. It recovers with the lever at IDLE and stalls again above the threshold (QRH: reduce thrust below it). |
| Engine 1(2) EGT overtemperature              | 72 Engine            | The EGT is higher in proportion to N1 (about 100 °C at climb thrust): ENG EGT OVER LIMIT at climb thrust.                                                                                                                                         |
| Engine 1(2) N1/N2 overspeed indication       | 72 Engine            | The N1 and N2 indications read 7 % / 6 % high: ENG N1 OVER LIMIT at takeoff thrust. The thrust is unchanged.                                                                                                                                      |
| Engine 1(2) high vibration                   | 77 Engine Indicating | N1 and N2 vibrations growing with the square of the rotor speed: above the SD advisory (6 / 4.3) at climb and cruise thrust, below it at reduced thrust and clearly below at idle. No ECAM alert (QRH procedure).                                 |

- **E/WD limits** (now in one shared place): EGT amber above 940 °C at TOGA/FLX, 905 °C at CLB/MCT, 725 °C without a rated limit or in the start; red above 975 °C, with a red mark at the highest EGT reached (it stays until the next engine start on the ground). N1 amber above its limit, red above 104 %; N2 red above 105 % with a red cross. These are the CFM56-5B TCDS values the user chose.
- **SD ENGINE page:** separate N1 and N2 vibrations, pulsing above the advisory.

## What you should see

| Situation                      | What you should see                                                                                                                                                                                                       |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stall in flight                | Amber <b>ENG 1 STALL</b> : -THR LEVER 1 ... IDLE, -ENG 1 PARAMETERS ... CHECK, -ENG 1 STALL PROC ... APPLY; SD ENGINE page. On the ground: -THR LEVER 1 ... IDLE, -ENG MASTER 1 ... OFF.                                  |
| A parameter above its limit    | Amber <b>ENG 1 EGT OVER LIMIT</b> (N1, N2 OVER LIMIT; the title names the parameter): -THR LVR 1 .. BELOW LIMIT; once the EGT went above 975 °C (or N1 / N2 above 105.8 %): -THR LEVER 1 ... IDLE, -ENG MASTER 1 ... OFF. |
| High vibration at climb thrust | VIB N1 6 or more and VIB N2 4.3 or more, pulsing green.                                                                                                                                                                   |

## Limitations and why

| Limitation                                                                                         | Why                                                                        | What to do                                               |
|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------|
| What each failure does (N1 loss, EGT and indication offsets, vibration levels) is a design choice. | The FCOM gives the symptoms, not values.                                   | -                                                        |
| The EGT limits are the CFM56-5B TCDS values.                                                       | The LEAP-1A limits are an open question; the FCOM text gives 950 / 915 °C. | -                                                        |
| Inhibitions: STALL in flight phases 3, 4, 5, 7, 8; OVER LIMIT in 4 and 8.                          | The FCOM figures.                                                          | Test the stall in the climb above 1 500 ft or in cruise. |
| A full MSFS restart is needed after an update.                                                     | The systems, FADEC and fbw WASM modules change.                            | Restart the sim after installing.                        |

How to use

- 1. Open flyPad > Failures > ATA 72 Engine or ATA 77 Engine Indicating and activate a failure at climb or cruise thrust.
- 2. Follow the ECAM (THR LEVER IDLE or below the limit); deactivate the failure after the test.

How to test

From the integration plan S8 T3 and S9 T1 (recorded OK on develop 2026-10-06). Setup: climbing through FL 150 at about 280 kt, A/THR ON at CLB, AP ON, both engines running.

| # | Do this                                                                                             | Expected result                                                                                                                                                                                                                                          |
|---|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Activate <b>Engine 1 compressor stall (above 60 % N1)</b> . Wait 20 s.                              | Amber ENG 1 STALL with its three lines; SD ENGINE page; N1 1 fluctuates and settles about 15 % below N1 2; EGT 1 rises by about 100 °C or more.                                                                                                          |
| 2 | Thrust lever 1 to IDLE, wait 20 s.                                                                  | Within about 10 s the alert goes; N1 1, EGT 1 and the vibration back to normal.                                                                                                                                                                          |
| 3 | Thrust lever 1 back to CL, wait 10 s; deactivate the failure.                                       | The stall comes back, then ends with the failure.                                                                                                                                                                                                        |
| 4 | At CL, activate <b>Engine 1 EGT overtemperature</b> and <b>Engine 1 high vibration</b> . Wait 30 s. | EGT 1 about 100 °C above EGT 2, amber above 905 °C with ENG 1 EGT OVER LIMIT and -THR LVR 1 .. BELOW LIMIT (above 975 °C: red, -THR LEVER 1 ... IDLE, -ENG MASTER 1 ... OFF, and a red mark stays); SD VIB N1 6 or more and VIB N2 4.3 or more, pulsing. |
| 5 | Thrust lever 1 to IDLE, wait 30 s; deactivate both failures, lever back to CL.                      | The alert goes; EGT 1 green; both vibrations steady green below the advisory (VIB N2 stayed above it at idle before the tuning). A red EGT mark stays.                                                                                                   |
| 6 | Optional (not sim-tested): at takeoff thrust, activate <b>Engine 1 N1/N2 overspeed indication</b> . | N1 1 and N2 1 read higher than engine 2 with the same thrust; ENG 1 N1 OVER LIMIT.                                                                                                                                                                       |

14. FADEC, thrust lever and reverser failures

**Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 73 Engine - Fuel And Control, ATA 78 Exhaust; thrust levers; E/WD, STATUS, SD | **Status:** Sim-tested by the user on develop: OK (2026-10-06 integration flight S8 T2, recorded: Thrust lever 1 resolvers (both), then Engine 2 FADEC channel A and B). The reverser fault test of that flight found no reverse thrust at all (the in-flight start hold clamped it): fixed, normal reverse on the ground data-checked OK 2026-10-07; the reverser failures were not re-tested after the fix. FADEC overheat, resolver disagree, reverser unlocked and pressurized: not sim-tested. Rust tests engine\_control\_failure.rs and hydraulic, native tests fadec\_failure\_inputs\_test and reverse\_throttle\_test, unit tests FadecReverserAlerts.spec.ts, A32NX\_FADEC.spec.ts.

What it is for

| flyPad failure (both engines / levers / reversers) | ATA | What the code does                                                                                                                                                                                                         |
|----------------------------------------------------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Engine 1(2) FADEC channel A / channel B            | 73  | One channel lost: the other takes control (ENG FADEC A (B) FAULT). Both lost: ENG FADEC FAULT, the engine data are lost (XX on the E/WD and SD), the A/THR disconnects, no reverser; the engine keeps following its lever. |
| Engine 1(2) FADEC overheat                         | 73  | ENG FADEC HI TEMP (ECU above 105 °C).                                                                                                                                                                                      |
| Thrust lever 1(2) resolvers (both)                 | 73  | The FADEC selects idle, CLB or frozen takeoff thrust by the FCOM rules: in flight it keeps CLB (or the takeoff thrust if the lever failed in TO/FLX) until the slats are out, then idle; idle on the ground; no reverser.  |

|                                                |            |                                                                                                                                |
|------------------------------------------------|------------|--------------------------------------------------------------------------------------------------------------------------------|
| Thrust lever 1(2) resolver disagree            | 73         | ENG THR LEVER DISAGREE: idle on the ground, CLB as the maximum in cruise, idle in approach.                                    |
| Reverser 1(2) fault (does not deploy)          | 78 Exhaust | The reverser stays stowed; its engine stays at reverse idle: ENG REVERSER FAULT.                                               |
| Reverser 1(2) unlocked (engine at idle)        | 78 Exhaust | The doors leave their stowed and locked position (no reverse thrust); the FADEC sets the engine at idle: ENG REVERSE UNLOCKED. |
| Reverser 1(2) pressurized (shutoff valve open) | 78 Exhaust | ENG REV PRESSURIZED.                                                                                                           |

- Reverse thrust stays at reverse idle until the reverser is fully deployed (FCOM DSC-70-70): a reverser that does not deploy no longer gives forward thrust with the lever in reverse. The hydraulic reversers report their status to the FADECs and stop deploying without a FADEC command.
- Fix: the in-flight start throttle hold (feature "Engine flameout ...") clamped the MSFS throttle at 0, which blocked all reverse thrust on the ground; the reverse throttle now reaches MSFS.

## What you should see

| Situation                                    | What you should see                                                                                                                                                                                                                                  |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| One FADEC channel failed                     | Amber <b>ENG 1 FADEC A FAULT</b> (B), crew awareness.                                                                                                                                                                                                |
| Both channels failed                         | <b>ENG 2 FADEC FAULT</b> : in flight -THR LEVER 2 ... IDLE, -ENG 2 PARAMETERS ... CHECK, .IF ABN ENG OPERATION: -ENG MASTER 2 ... OFF (on the ground THR LVR 2 NOT ABV IDLE); engine 2 N1, EGT, N2, FF XX; A/THR OFF; STATUS THR LVR 2 NOT ABV IDLE. |
| FADEC overheat                               | <b>ENG 1 FADEC HI TEMP</b> , FADEC OVHT; in flight -ENG 1 PARAMETERS ... CHECK, .IF ABN ENG OPERATION: -THR LEVER 1 ... IDLE, -ENG MASTER 1 ... OFF; on the ground also -ENG MODE SEL ... NORM, -FADEC GND PWR ... CHK OFF.                          |
| Thrust lever resolvers lost, in flight at CL | <b>ENG 1 THR LEVER FAULT</b> with -A/THR ... KEEP ON (A/THR engaged) or ENG 1 HI PWR IN MAN THR; LAND ASAP amber; STATUS WHEN SLATS OUT: ENG 1 AT IDLE, INOP SYS REVERSER 1, ENG 1 THR. With the slats out: ENG 1 AT IDLE.                           |
| Resolver disagree                            | <b>ENG 1 THR LEVER DISAGREE</b> : on the ground ENG 1 IDLE POWER ONLY; in cruise AVAIL MAX POWER : CLB with -A/THR ... KEEP ON (or ON); in approach ENG 1 AT IDLE. STATUS ENG 1 AVAIL MAX PWR:CLB, GND ENG 1 MAX PWR: IDLE, INOP SYS ENG 1 THR.      |
| Reverser fault                               | Amber <b>ENG 1 REVERSER FAULT</b> (title only); STATUS INOP SYS REVERSER 1; with the levers in reverse, REV only on the other engine.                                                                                                                |
| Reverser unlocked in flight                  | <b>ENG 1 REVERSE UNLOCKED</b> : ENG 1 AT IDLE, -THR LEVER 1 ... IDLE, -MAX SPEED ... 300/.78, .IF BUFFET: -MAX SPEED ... 240 KT, -ENG MASTER 1 ... OFF; STATUS MAX SPEED 300/.78.                                                                    |
| Reverser pressurized                         | <b>ENG 1 REV PRESSURIZED</b> : in flight -THR LEVER 1 ... IDLE; on the ground THR LVR 1 NOT ABV IDLE.                                                                                                                                                |

## Limitations and why

| Limitation | Why | What to do |
|------------|-----|------------|
|------------|-----|------------|

|                                                                                                                                                                                                                                                                                                                    |                                                                                                       |                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-----------------------------------|
| Inhibitions (FCOM figures):<br>FADEC A(B) FAULT and FADEC HI<br>TEMP phases 3, 4, 5, 7, 8; FADEC<br>FAULT 4, 5, 7, 8; REV<br>PRESSURIZED 4, 5, 8; REVERSE<br>UNLOCKED 4, 5, 8; REVERSER<br>FAULT 3, 4, 5; THR LEVER FAULT<br>/ DISAGREE 4, 5. Some are not<br>inhibited when the FADEC sets<br>the engine at idle. | The FCOM gives them as figures.                                                                       | Test on the ground or in cruise.  |
| The flyPad reverser fault gives<br>the title only.                                                                                                                                                                                                                                                                 | It is not a position fault: the FCOM lines are for a<br>position fault with the reverser pressurized. | -                                 |
| A reverser that really deploys in<br>flight is not modelled.                                                                                                                                                                                                                                                       | The unlocked doors stay in the stowed position.                                                       | -                                 |
| FADEC OVHT is always shown<br>with ENG FADEC HI TEMP.                                                                                                                                                                                                                                                              | Design choice: the flyPad overheat is an ECU above<br>105 °C.                                         | -                                 |
| A full MSFS restart is needed<br>after an update.                                                                                                                                                                                                                                                                  | The systems and fbw WASM modules change.                                                              | Restart the sim after installing. |

## How to use

1. Open flyPad > Failures > ATA 73 Engine - Fuel And Control or ATA 78 Exhaust and activate a failure.
2. Follow the ECAM and STATUS; deactivate the failure after the test.

## How to test

From the integration plan S8 T1-T2 and the retest card T2. Check the flyPad throttle calibration (reverse and idle detents) first.

| # | Do this                                                                                                                                  | Expected result                                                                                                                                                                |
|---|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | On the runway, both engines at idle, parking brake ON.<br>Activate <b>Reverser 1 fault (does not deploy)</b> .                           | Amber ENG 1 REVERSER FAULT; STATUS INOP SYS<br>REVERSER 1.                                                                                                                     |
| 2 | Both thrust levers to full reverse for 5 s, then IDLE. Deactivate<br>the failure.                                                        | REV green on engine 2 only; engine 2 N1 rises, engine 1<br>N1 stays near idle; at IDLE the reverser stows, no other<br>caution.                                                |
| 3 | After a landing (no failure): both levers to full reverse until 70<br>kt, then IDLE.                                                     | REV green on both engines, N1 well above idle, the<br>aircraft decelerates; no forward thrust jump at IDLE (OK<br>2026-10-07).                                                 |
| 4 | FL 100, 220 kt, flaps 0, AP and A/THR ON, levers in CL. Activate<br><b>Thrust lever 1 resolves (both)</b> , then thrust lever 1 to IDLE. | ENG 1 THR LEVER FAULT with -A/THR ... KEEP ON and<br>LAND ASAP; STATUS WHEN SLATS OUT: ENG 1 AT IDLE,<br>INOP SYS REVERSER 1, ENG 1 THR; engine 1 N1 does<br>not drop to idle. |
| 5 | FLAPS 1.                                                                                                                                 | Engine 1 N1 goes to idle and stays there.                                                                                                                                      |
| 6 | Activate <b>Engine 2 FADEC channel A</b> , then <b>Engine 2 FADEC<br/>channel B</b> .                                                    | ENG 2 FADEC FAULT; engine 2 N1, EGT, N2, FF XX on the<br>E/WD and the SD ENGINE page; the A/THR disconnects;<br>thrust lever 2 still changes engine 2 thrust.                  |
| 7 | Deactivate all failures, FLAPS 0, lever 1 to CL, A/THR ON.                                                                               | All alerts clear.                                                                                                                                                              |

## 15. T.O CONFIG NORMAL and FUEL WING TK LO LVL fixes

**Aircraft:** A32NX | **Where:** ECAM control panel T.O CONFIG key; E/WD, STATUS | **Status:** Not sim-tested. Unit tests ToConfigNormal.spec.ts, WingTankLowLevel.spec.ts.

### What it is for

Two deviations from the A320 FCOM found by the FCOM gap audit (the rest of that batch is on `core/fcom-audit-fixes`). They use the wing tank pump alerts of this branch.

- T.O CONFIG NORMAL** is no longer given with a FUEL L(R) TK PUMP 1+2 LO PR alert (FCOM DSC-31-15: the T.O CONFIG test triggers it, and NORMAL needs no such alert).
- FUEL L(R) WING TK LO LVL** (inner tank below 750 kg for 30 s) asks for -FUEL MODE SEL ... MAN only if the centre tank is not empty, then .IF NO FUEL LEAK AND FUEL IMBALANCE: -FUEL X FEED ... ON, -L TK PUMP 1 / 2 ... OFF. It now gives its STATUS: INOP SYS L (R) TK PUMPS and, with fuel in the centre tank, CTR TK FEED: MAN ONLY (FCOM PRO-ABN-FUEL).

### Limitations and why

| Limitation                                                             | Why                                          | What to do                                                                            |
|------------------------------------------------------------------------|----------------------------------------------|---------------------------------------------------------------------------------------|
| FUEL WING TK LO LVL is inhibited in flight phases 3, 4, 5, 7, 8 and 9. | FBW inhibition, unchanged.                   | Test at the gate or in cruise.                                                        |
| A low level in one wing only needs a fuel tool.                        | The flyPad fuel page fills both wings alike. | Set the tank quantity with a SimConnect variable tool (FUEL TANK LEFT MAIN QUANTITY). |

### How to use

- Nothing to set: the alerts follow the FCOM by themselves.
- Press the T.O CONFIG key before takeoff as usual; follow FUEL WING TK LO LVL and its STATUS if it comes.

### How to test

| # | Do this                                                                                                                                         | Expected result                                                                                                                             |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | A32NX at the gate, engines running, all FUEL pumps ON.<br>flyPad > Failures > ATA 28 Fuel: activate <b>L TK pump 1</b> and <b>L TK pump 2</b> . | FUEL L TK PUMP 1+2 LO PR.                                                                                                                   |
| 2 | Press the T.O CONFIG key (ECAM control panel).                                                                                                  | No T.O CONFIG NORMAL in the T.O memo. Deactivate the failures: T.O CONFIG gives T.O CONFIG NORMAL again (with a takeoff configuration set). |
| 3 | Centre tank empty; set the left inner tank to about 600 kg with a SimConnect tool (right inner tank above 750 kg). Wait 30 s.                   | FUEL L WING TK LO LVL without the -FUEL MODE SEL ... MAN line; STATUS INOP SYS L TK PUMPS.                                                  |
| 4 | Put fuel in the centre tank (FUEL MODE SEL at AUTO).                                                                                            | The -FUEL MODE SEL ... MAN line shows; STATUS also CTR TK FEED: MAN ONLY.                                                                   |

## 16. ENG THR LEVER ABV IDLE and ENG SAT ABOVE FLEX TEMP

**Aircraft:** A32NX | **Where:** E/WD; thrust levers; MCDU PERF TAKE OFF page (FLEX TO TEMP) | **Status:** SAT ABOVE FLEX TEMP: tested in the sim by the user: OK, data-confirmed (2026-10-08, on the feature branch). THR LEVER ABV IDLE: not sim-tested (the first test card was wrong, the test after a real landing is not run yet). Unit tests EngineThrustSettingAlerts.spec.ts.

What it is for

Two thrust setting alerts of the A320 FCOM (PRO-ABN-ENG) that the A32NX did not have. They need no failure: they come from the thrust levers and the FLEX TEMP.

- **ENG 1(2) THR LEVER ABV IDLE** (red, MASTER WARN, repetitive "RETARD-RETARD"; FCOM I.81632-81666): during the landing roll (FWC phases 8 and 9), one thrust lever is above idle while the other one is in reverse, or stays above idle after the other one came out of reverse. Line: -THR LEVER 1(2) ... IDLE. The voice is heard above 40 kt and stops when all levers are at IDLE or REVERSE.
- **ENG SAT ABOVE FLEX TEMP** (amber, MASTER CAUT; FCOM I.80984-81000): before take-off (phase 2 only) the outside air temperature is above the FLEX TEMP entered in the MCDU. Line: T.O DATA ... CHECK. No FLEX TEMP entered gives no alert.

Limitations and why

| Limitation                                               | Why                                                                                                                                          | What to do               |
|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| THR LEVER ABV IDLE only after a real landing.            | The FWC goes to phase 9 only after a touchdown or a take-off aborted above 80 kt; a take-off power cut at a standstill goes back to phase 2. | Test it after a landing. |
| "At reverser deselection" has no time limit in the FCOM. | Design choice: the memory of a reverse selection is kept until both levers are at idle or the landing roll ends.                             | -                        |

How to use

1. Nothing to set. Before take-off, enter a FLEX TEMP above the outside temperature (or none for TOGA).
2. After landing, bring both thrust levers to REVERSE or IDLE together; follow the RETARD call if it comes.

How to test

| # | Do this                                                                                                                                                     | Expected result                                                                                    |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| 1 | A32NX at the gate, engines running, FWC phase 2. MCDU PERF TAKE OFF: enter a FLEX TO TEMP 10 °C below the outside temperature (SAT).                        | Within a second: amber ENG SAT ABOVE FLEX TEMP with T.O DATA ... CHECK, single chime, MASTER CAUT. |
| 2 | Enter a FLEX TO TEMP of 60.                                                                                                                                 | The caution goes; the T.O memo is back.                                                            |
| 3 | After a landing, stopped on the runway or a taxiway (phase 9), both levers at IDLE, PARKING BRK ON: lever 2 to REVERSE IDLE, then lever 1 to the CL detent. | Red ENG 1 THR LEVER ABV IDLE with -THR LEVER 1 ... IDLE, MASTER WARN.                              |
| 4 | Lever 2 forward to IDLE (lever 1 stays at CL), then lever 1 to IDLE.                                                                                        | The warning stays while lever 1 is at CL, and goes once it is at IDLE.                             |

17. Engine fan blocked failure and ENG LOW N1

**Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 72; ENG MASTER and ENG MODE; E/WD | **Status:** Not sim-tested (test T4 of the engine alerts card not run yet). Unit tests EngineLowN1Alerts.spec.ts, a320.spec.ts; native test fan\_blocked\_start\_test.cpp; Rust failure type.

What it is for

The A320 FCOM alert **ENG 1(2) LOW N1** (on ground, PRO-ABN-ENG I.80343-80363) is for a start in which the fan does not turn (a blocked fan, for example ice). The sim cannot produce that by itself, so the branch adds a failure for it.

- New flyPad failures, ATA 72: **Engine 1 fan blocked (no N1 rotation at start)** and **Engine 2 fan blocked (no N1 rotation at start)** (ids 72020, 72021). During a start on the ground the FADEC holds the N1 at 0 and the core hangs below idle (about 41 % N2).
- ENG 1(2) LOW N1 (amber, MASTER CAUT, single chime): N2 at or above 35 % with N1 below 2 % for 2 s, on the ground; lines .IF CONFIRMED:, -THR LEVER 1(2) ... IDLE (only with the lever above idle), -ENG MASTER 1(2) ... OFF. It stays until the ENG MASTER is set OFF; inhibited in flight phases 4 to 9.

## Limitations and why

| Limitation                                              | Why                                                                     | What to do                              |
|---------------------------------------------------------|-------------------------------------------------------------------------|-----------------------------------------|
| The 35 % N2, 2 % N1 and 2 s values are not in the FCOM. | Design choices for "no N1 rotation".                                    | -                                       |
| ENG START FAULT may follow.                             | The FADEC aborts an automatic start whose core hangs, about 20 s later. | Both alerts are expected; LOW N1 stays. |
| A full MSFS restart is needed after an update.          | The systems and FADEC WASM modules change.                              | Restart the sim after installing.       |

## How to use

1. flyPad > Failures > ATA 72 Engine: activate "Engine 1 (2) fan blocked (no N1 rotation at start)" before the engine start.
2. Start the engine as usual; follow ENG LOW N1 (ENG MASTER OFF). Clear the failure before the next start.

## How to test

| # | Do this                                                                                                                                                                            | Expected result                                                                                                                                                                                        |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | A32NX on the ground, PARKING BRK ON, APU BLEED ON, engine 1 OFF, both levers at IDLE. flyPad > Failures > ATA 72: activate <b>Engine 1 fan blocked (no N1 rotation at start)</b> . | No alert yet.                                                                                                                                                                                          |
| 2 | ENG MODE selector IGN/START, ENG MASTER 1 ON. Watch engine 1 for 60 s.                                                                                                             | N2 rises, N1 stays at 0. Once N2 is above 35 % for 2 s: amber ENG 1 LOW N1 with .IF CONFIRMED: and -ENG MASTER 1 ... OFF, single chime, MASTER CAUT. N2 hangs near 41 %; ENG 1 START FAULT may follow. |
| 3 | ENG MASTER 1 OFF.                                                                                                                                                                  | ENG 1 LOW N1 goes (and START FAULT with it).                                                                                                                                                           |
| 4 | Clear the failure, wait 30 s, ENG MASTER 1 ON.                                                                                                                                     | A normal start: N1 rises with N2, no LOW N1.                                                                                                                                                           |